Rakibul Islam Rakib

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PROFILE

Hardware and electronics engineer with 6+ years of hands-on robotics experience and an international track record, including a 6th-place, full-score finish at the World Robot Olympiad 2025 International Round. I design complete systems end to end, from schematic capture and custom PCBs to embedded firmware, closed-loop control, and mechanical integration, and I have delivered quadruped robots, autonomous platforms, drones, and a 200 m-rated underwater vehicle. I lead teams, train engineers, and turn ambitious ideas into working hardware that performs under real-world conditions.

TECHNICAL SKILLS

Embedded Systems: Bare-metal and Arduino-framework firmware in C/C++ for ESP32 and AVR microcontrollers; peripheral interfacing over I^2C, SPI, UART, PWM, and ADC; sensor integration, signal conditioning, and multi-sensor data acquisition

Hardware & PCB Design: Schematic capture and multi-layer PCB layout in KiCad and EasyEDA; in-house PCB fabrication; analog/digital circuit design; MOSFET power switching and fail-safe kill circuits; power distribution and battery management for robotic systems; board bring-up and hardware debugging

Robotics & Control Systems: Forward/inverse kinematics for legged (quadruped) locomotion; closed-loop feedback control and PID tuning; autonomous navigation and state-machine logic; UAV flight-controller configuration and tuning; brushless thruster and actuator drive systems

Mechanical & Prototyping: Parametric CAD in Onshape; FDM 3D printing and rapid prototyping; pressure-hull design and waterproof sealing for subsea systems

Software & Tools: C/C++, Python; PlatformIO, Arduino IDE; Git/GitHub version control; web development and deployment

Leadership: Team management, technical training and mentoring, event planning and execution

Languages: Bangla (fluent), English (fluent)

EXPERIENCE

Zero Competitions Dhaka, Bangladesh
IT Executive 2026 – Present

- Own end-to-end IT operations for the organization, from web infrastructure to event-day technical systems.
- Develop, deploy, and maintain the official website, including domain/DNS management, hosting, and performance optimization.
- Build and manage online registration, submission, and result-publishing workflows for national-level competitions.
- Administer digital tools, accounts, and data security practices across the team; provide technical support to organizers.

BRACU Duburi Dhaka, Bangladesh
Member, Electronics and Mechanical Team 2025 – 2026

- Developed custom in-house underwater thrusters with performance comparable to the Blue Robotics T200.
- Designed the full electronics architecture of the DUBO MINI vehicle, including a MOSFET-based kill switch.
- Designed torpedo and dropper attachment mechanisms.
- Engineered hull design and waterproofing rated to a depth of 200 m.
- Contributed to the team achieving an 8th-place worldwide ranking in 2026.

World Robot Olympiad – Bangladesh National Team (Team LazyGo) International
Team Member, Hardware and Electronics 2025

- Designed and built the robot's complete hardware and electronics system for international competition.
- Placed 6th worldwide, achieving the maximum possible score in the technical challenge.

World Robot Olympiad – Bangladesh National Team International
Team Lead 2024

- Led the national team through design, build, and competition, representing Bangladesh internationally.
- Achieved a 19th-place world ranking.

Robodemy Dhaka, Bangladesh
Associate Trainer and R&D Developer 2024 – 2025

- Taught robotics and electronics to students and mentored them through hands-on projects.
- Developed two inverse-kinematics-based quadruped robots, including complete hardware architecture and firmware.

DRMC Science Club Dhaka, Bangladesh
Vice President, Executive Panel 2024 – 2025

- Led the planning and execution of an international science and technology festival in April 2025.
- Managed event coordination, team organization, and program execution.

Tech Topia Dhaka, Bangladesh
Tech Envoy and R&D Team Member 2022 – 2024

- Organized and supported technology workshops and programs as a volunteer Tech Envoy.
- Contributed to the R&D team in designing and improving technical projects and learning materials.

EDUCATION

BRAC University Dhaka, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering Spring 2026 – Present

Dhaka Residential Model College Dhaka, Bangladesh
Higher Secondary Certificate (HSC), Dhaka Board. GPA: 5.00/5.00 2025

Annada Government High School Bangladesh
Secondary School Certificate (SSC), Cumilla Board. GPA: 5.00/5.00 2023

HONORS & AWARDS

World Robot Olympiad, International Round – 6th place worldwide (full score) 2025
World Robot Olympiad, Open Championship National Selection Round – Champion (Gold Medalist) 2025
World Robot Olympiad, National Round – Champion (Gold Medalist) 2025
World Robot Olympiad, Regional Round – Champion 2025

World Robot Olympiad, International Round – Participant, 19th place worldwide	2024
World Robot Olympiad, National Round – Champion (Gold Medalist)	2024
World Robot Olympiad, National Round – Winner (Silver Medalist)	2023
Fibonacci National Round Bangladesh – Champion, Micromouse	2026
Fibonacci National Round Bangladesh – 1st Runner-up, Micromouse	2026
DRMC Science Carnival – 1st Runner-up, Drone Race	2026
Robotronics Fest – Runner-up, Micromouse	2025
Robotronics Fest – Runner-up, Line Following Robot	2025
15th DRMC National Science Carnival – Champion, Mini Soccer	2025
15th DRMC National Science Carnival – Champion, Fire Fighter Robot	2025
UIU CSE Fest – 1st Runner-up, Line Following Robot	2025
4th International Robotech Olympiad – Champion, Line Following Robot	2024
EWU National RoboFest – 1st Runner-up, Line Following Robot	2024
Codeavour Bangladesh – Winner	2024
Init 4.0 – Runner-up, Line Following Robot	
Green University EEE Day – 2nd place, Line Following Robot	
OSCILLON, IEEE AUST Student Branch – 2nd place, Line Following Robot	
Technomize 1.0, KUET – 1st Runner-up, Line Following Robot	
Calibration 2.0 – Champion, Micromouse	
Calibration 2.0 – 1st Runner-up, Micromouse	
Calibration 2.0 – 1st Runner-up, Line Following Robot	
Calibration 2.0 – 2nd Runner-up, Line Following Robot	